



Positively Staggered Vertically Offset Propellers in Tandem

Jielong Cai¹ and Sidaard Gunasekaran²

University of Dayton, Dayton, OH, 45469

Michael OL³

Folderol, LLC

We continue our study on the aerodynamic investigation of two propellers mounted partially overlapping and in tandem configurations. In the previous study, the rear propeller was placed above the plane of the front propeller (negative stagger). The current study placed the rear propeller beneath the plane of the front propeller (positive stagger). Different vertical and horizontal spacings, incidence angles with respect to the freestream, and advance ratios were tested experimentally at the University of Dayton Low Speed Wind Tunnel. Thrust and torque were measured separately on each propeller. A significant increment in the power required of the tandem propeller system was observed when compared to the results in the previous study under negative staggered conditions. A higher loss in system efficiency at small forward flight speeds is also observed due to propeller wake interaction. The advance ratio at which this performance loss occurs is affected by the propeller spacing and incidence angle.

I. Nomenclature

α_{in}	= Propeller Inflow Angle, (deg)	n	= Propeller Rotational Speed, (RPS)
C_T	= Propeller Thrust coefficient; $C_T = \frac{T}{\rho n^2 D^4}$	P	= Power, (W)
C_P	= Power Coefficient; $C_P = 2\pi C_Q = \frac{P}{\rho n^3 D^5}$	P_s	= Single Propeller Power Required, (W)
C_Q	= Propeller Torque Coefficient; $C_Q = \frac{Q}{\rho n^2 D^5}$	P_t	= Tandem Propeller Power Required, (W)
C_{Qx}	= Propeller Pitching Moment Coefficient	γ	= Propeller Pitch, (m)
C_{Qy}	= Propeller Rolling Moment Coefficient	Q	= Torque, (Nm)
D	= Propeller Diameter, (m)	R	= Propeller Radius, (m)
d	= Distance Between Propeller Hubs, (m)	T	= Thrust, (N)
J	= Propeller Advance Ratio; $J = \frac{V_\infty}{nD}$	T_s	= Single Propeller Thrust, (N)
		T_t	= Tandem Propeller Thrust, (N)
		V_∞	= Freestream Velocity, (m/s)
		ρ	= Air Density, (kg/m ³)
		θ	= Propeller-plane Incidence Angle, (deg)

II. Introduction

The increasing usage of propeller-driven aircraft, including unmanned aerial vehicles (UAVs) and eVTOL air taxis, broadens the classical helicopter problem of edgewise flight. One common method to increase the payload of the eVTOL vehicle is to increase the number of propellers instead of using a larger rotor. A higher

¹ Graduate student, Mechanical and Aerospace Department, AIAA Member, caij03@udayton.edu

² Associate professor, Mechanical and Aerospace Department, AIAA Member, gunasekarans1@udayton.edu

³ AIAA Associate Fellow, riven.pithos@gmail.com

number of propellers also helps achieve a lower disk loading, thereby increasing efficiency and range [1]. However, eVTOL vehicles commonly face the constraint of minimizing their overall size or ground footprint to facilitate operations on small spaces such as rooftops or helipads. Coaxial rotors are considered a solution for reducing the footprint, a concept occasionally employed in helicopters to minimize torque imbalance [1]. However, an average performance loss of 15 ~ 25% was reported when compared to a single rotor in hover, as the lower rotor operates constantly in the wake of the upper rotor [2-5]. This loss is reduced to 9 ~ 15% in high-speed forward flight [2,6,7].

An alternative to a coaxial rotor is to mount the propellers in close proximity within the same plane, as seen in the VoloCity air taxi [8] in Figure 1. The loss in hover performance is negligible when the propellers are placed over $1.1 d/D$ apart, between rotor hubs [5, 10–12, 17]. However, a performance loss of 11 ~ 18% in low-speed forward flight is observed when $d/D < 3$, due to rotor interaction in forward flight adversely affecting the downstream rotor [2, 6, 7, 11-18]. Moreover, an overall thrust difference between the front and rear rotor can be up to ~40% based on flight test data in Ref. [13] and Ref. [16-17].

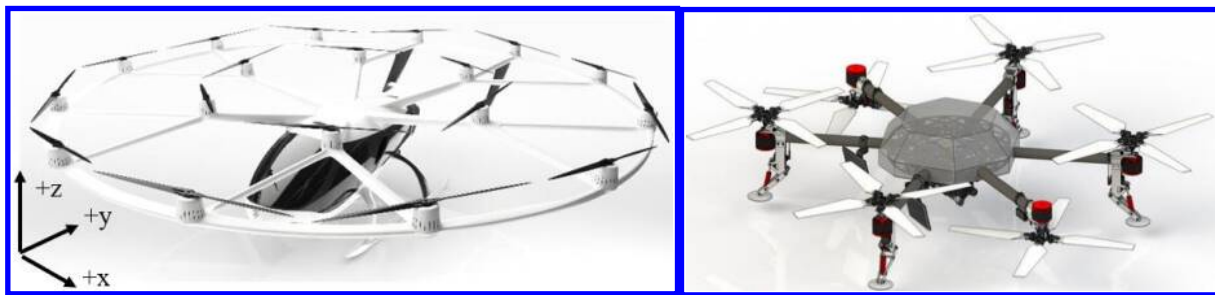


Figure 1. VoloCity air taxi [8] (left) and JAXA Mars exploration helicopter [9] (right).

However, large spacing vitiates compactness (Figure 1). For a standard square quadcopter, changing the flight configuration can also reduce the performance loss in forward flight. Based on the wind tunnel experiment [17,18] and flight test [16,18], flying the quadcopter in a diamond/plus configuration results in a ~10% performance increment when compared to the cross configuration in forward flight. This is due to the reduction in the rotor-rotor interaction, as only two rotors are placed in tandem, resulting in only one rear rotor experiencing a significant performance loss [18].

An alternative solution is to place the rotors in tandem, with a vertical displacement, such as the JAXA Mars exploration helicopter [9] (Figure 1). Based on flow visualization studies and force-based testing, Ref. [13] suggests that the reduction of the aft rotor's performance is due to the changes in its inflow angle caused by the front rotor, which agrees with the Fast Multirotor Performance Prediction Model (FMPP) developed by Tsaltas [14]. Ref. [7] also concludes that placing the front rotor below the rear rotor reduces the influence on the rear rotor by the front rotor.

For hovering performance of tandem overlapping rotors, placing the upper rear rotor more than $0.12 d_z/D$ above the front rotor results in a minimum impact on the overall performance of both rotors, while overlapping the rotor at $d_y/D > 0.8$ also has a minimum impact [5,18-20]. At $d_y/D < 0.8$, the effect of the rotor overlap is almost proportional to the overlapping rotor disk area, which also holds true in forward flight [19]. However, the vertical spacing of the tandem rotor for large vehicles is usually limited to $d_z/D < 0.15$ due to physical limitations, as a further vertical placement will result in structural disbenefit [5, 13].

In our previous study [21,22], the performance of a tandem overlapping propeller pair was studied with the rear propeller placed above the front (negatively staggered, as shown in Figure 2 (a)). Increasing the vertical spacing between the propellers results in a decrease in power required at constant thrust at advance ratio $J > 0.2$;

in other words, an increase in efficiency in forward flight. There were no discernible further improvements when placing the rotors more than half the diameter vertically apart. The current study investigates the tandem pair, with the rear propeller beneath the front, which also resembles the reverse flight direction of the orientation in our previous study [21,22] (positively staggered as shown in Figure 2 (b)). This provides a full parametric study of the compact tandem propeller system aiming to provide a better design guideline for multi-propeller aircraft.

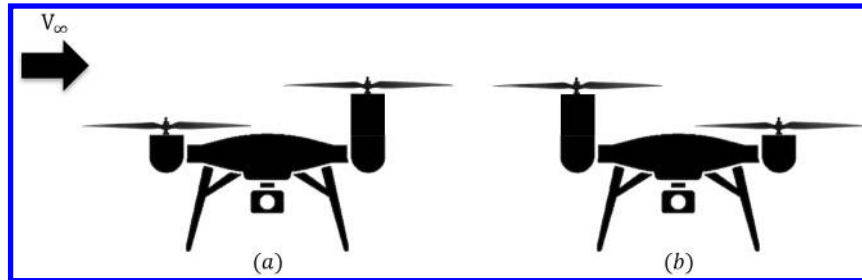


Figure 2. Schematic showing (a) negatively staggered and (b) positively staggered propeller orientations

III. Experimental Setup

A) Data Acquisition

Experiments were conducted at the University of Dayton Low-Speed Wind Tunnel (UD-LSWT) in the open jet test section. The wind tunnel has an inlet contraction ratio of 16:1, a 762 mm x 762 mm testing section inlet, and a 1.37 m x 1.37 m collector. The freestream range is from 1.5 m/s to 20 m/s, resulting in an advance ratio, J , between 0 and 0.5, with a turbulence intensity of 0.1% measured by a hot wire anemometer.

A sketch of the experimental setup is shown in Figure 3. The KDE CF-125 [23] propeller is paired with a KDE 2814-XF-515 brushless motor and Hobbywing Platinum V4 electronic speed controller operating in governor mode for better rotational speed control. The propeller and motor assembly are attached to an ATI Industrial Automation Mini-40 six-component force/torque transducer [24]. The transducer has a maximum of 40N on its x and y-axis, 120N on its z-axis, and a maximum torque of 2 Nm for all axes. Thrust is the axial-force along the z-axis of the sensor. The motor torque is the moment of the z-axis, while the propeller pitching moment is the torque on the x-axis, and the propeller rolling moment is the y-axis torque.

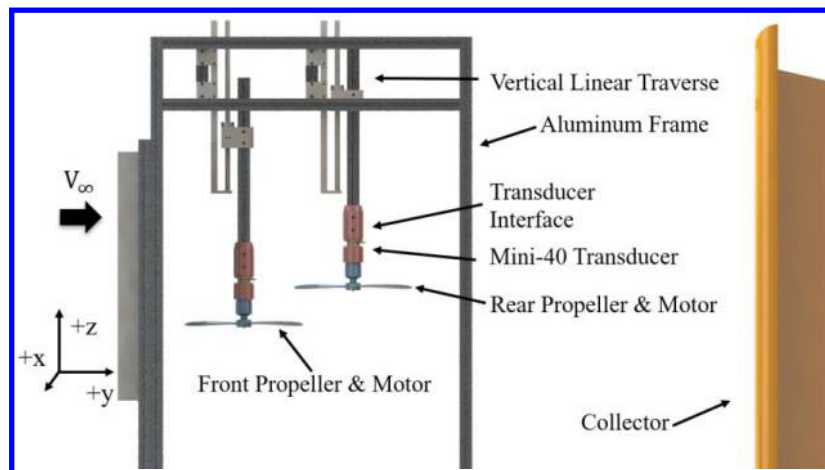


Figure 3. Schematic of the experimental setup at the UD-LSWT

The transducer is mounted on a 3D-printed interface, bolted to an 80-20 aluminum extrusion frame on a Fuyu FSL 40 linear traverse. The linear traverse is placed at the centerline of the test section. It changes the vertical distance (z) between the propellers. The propellers can be tilted with respect to the tunnel freestream. To change

the downstream distance (y), the location of the front rotor is fixed, while the rear propeller is moved back or forth. The entire assembly is mounted to a 1.22 x 0.92 x 2.44 meter aluminum frame, secured to the lab floor, the inlet of the wind tunnel, and the ceiling of the wind tunnel plenum. Two 10 mW lasers and two Thorlabs PDA10A Photodiodes are used to measure the rotational speed of each propeller.

The data was sampled at the rate of 1000 Hz for a 15-second duration at each data point. Two trials were run in each case to ensure data repeatability. To minimize sensor drift, tare values with the propeller powered off were taken before and after each test. The average of the two tares was subtracted from the respective thrust and torque readings. If the difference between the before and after tare value was greater than 0.1N of force on any axis, the experimental data was rejected. A Finite Impulse Response (FIR) band-stop filter [25] was applied to the data to filter out the vibration frequency caused by the propeller rotation, and the natural frequency of the aluminum test frame. Tap tests on the test stand with the propeller powered off, showed a structural natural frequency of 105Hz. With the propeller thrust acting on the structure during the wind-on experiment, the structure does not show significant vibration up to the highest propeller rotational speed tested. The average uncertainty in the force measurement after filtration is $\pm 8.6\%$ across all advance ratios tested, while the average uncertainty in torque is $\pm 7.4\%$. However, the results presented show excellent repeatability from both experimental runs.

B) Test Matrix

The test matrix is shown in Table 1. The KDE-CF 125 propeller was selected based on its popularity in modern drones. For relevance to actual n-copter drones, the propeller rotational speeds are set such that there is no net pitching moment, or moment about the X-axis in Figure 4; this is akin to trimming a drone in forward flight in pitch [15,16]. As the propeller will generate a pitch-up moment in edgewise flight, based on the result from our previous study [19,20], the rear propeller must generate a higher thrust to balance the pitch for the lateral stability of the drone. This is shown in the schematic in Figure 4. As the propeller is fixed-pitch, it must operate at a higher rotational speed to achieve trim.

Table 1. Test Matrix

Propeller	Rotational Speed (RPS)	Peak Reynolds Number (x1000)	Advance Ratio	Incidence Angle (Deg)	Vertical Placement d_z/D	Downstream Placement d_y/D
KDE-CF125	Front: 67 and 75 Rear: 67 - 128	180	0 - 0.8	75, 90	-0.2, -0.4, -0.6, -0.8	0.875, 1.0, 1.25

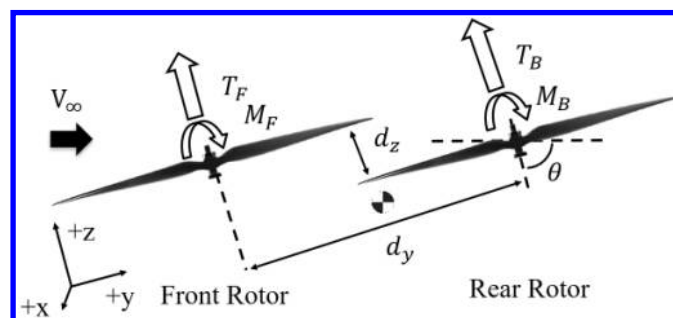


Figure 4 Nomenclature for tandem propellers.

Based on the sensor loading limit and the propeller balance, the front propeller rotational speed is selected at 67 and 75 RPS, resulting in a rear propeller rotational speed between 67 and 128 RPS, depending on the advance ratio. The peak Reynolds number is calculated at 75% blade span. To achieve different advance ratios, the

propeller RPS is fixed, while the wind tunnel freestream velocity is varied. 75 and 90 degree incidence angles (θ) were chosen, representing realistic operational range for drones in transition to forward flight and in forward flight. The vertical and downstream propeller placement is calculated based on the distance of the propeller hub, shown in Figure 4, while the value of each is between -0.2 to -0.8 (the rear propeller is beneath the front propeller) and 0.875 to 1.25, respectively.

C) Smoke Flow Visualization

Smoke flow visualization was performed with the front propeller rotating at 67 RPS. A smoke wand, with a diameter of 0.5 inches, was placed at 0.5 D upstream of the propeller at the inlet of the testing section. A 300-mW, 532 nm continuous laser was used as a light source. The advance ratio of the front propeller for smoke flow visualization was between 0.1 and 0.7.

IV. Results

A) Tandem Propeller Configuration at $\theta = 90^\circ$

To trim the tandem propeller pair in cumulative pitching moment, the rear propeller rotational speed is adjusted such that $C_{Qx\ norm}$ less than 5% for each advance ratio. $C_{Qx\ norm}$ is defined as:

$$C_{Qx\ norm} = \frac{C_{Qx\ Front} + C_{Qx\ Back} + (T_{Front} - T_{Back}) * 0.5d_y}{C_{Qx\ Front} + C_{Qx\ Back}} \approx 0 \quad (1)$$

Two front propeller rotational speeds are selected to check the repeatability of the experiment. As the overall Reynolds number effect is relatively small, the result for the front propeller at 4000 RPM (67 RPS) is discussed for all tandem configurations.

The resulting ratio of rotational speeds of the rear propeller (n_r) to the front (n_f) for $d_y = 1.25$ and $d_y = 1.0$ are shown in Figure 5, vs. advance ratio. The worst performance case from our previous study [21] is also shown in the plot (marked in red) for comparison.

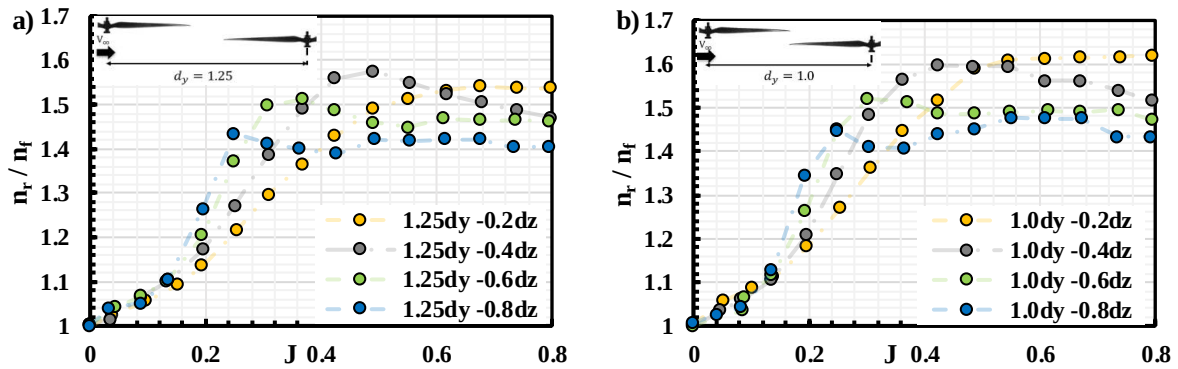


Figure 5. The ratio of the rear (n_r) to front ($n_f = 67$ RPS) propeller rotational speed, $\theta = 90^\circ$, for a) $d_y = 1.25$ and b) $d_y = 1.0$.

In both cases, trimmed n_f/n_r increases with J . For $J < 0.15$, n_f/n_r is independent of d_z . A sudden increase in n_f/n_r is observed between $0.2 < J < 0.6$, as a result of the propeller wake interaction between the front and the rear propeller. This phenomenon is observed at lower J for a lower d_z (larger propeller vertical spacing) as the wake of the front propeller will affect the rear propeller at a smaller wake skew angle for larger vertical spacing. At $J > 0.6$, the rotational speed for the rear propeller no longer increases with J . Instead, n_r/n_f

merges towards the $d_z = 0.0$ cases except for $d_z = -0.2$, indicating a reduction in the propeller wake interaction at higher J . With the increase in d_z towards $d_z = 0.0$, the influence of the front propeller on the rear propeller increases, so the rotational speed of the rear propeller has to increase to generate the required thrust to trim, as observed in our previous study [21]. When reducing d_y , due to the reduction in the moment arm between the propellers, as well as the increase in the propeller interaction, the rear propeller rotational speed increases even further at the same d_z .

A. 1) Effect of Front and Rear Propeller Interaction at $\theta = 90^\circ$

$\theta = 90^\circ$ resembles the hovering condition and the transition to forward flight. The front propeller will be discussed first. The thrust coefficient (C_T) at different advance ratios for all d_y and the standalone propeller which serves as a baseline are shown in Figure 6. a, b, and c. C_T for all cases is nearly constant for $J < 0.2$, and then rises linearly, which agrees with the trend for the standalone propeller. Unlike the $d_z \geq 0.0$ cases, where a positive effect of the rear propeller on the front is observed resulting in a higher C_T for the front rotor, the difference between different d_z cases on C_T is almost negligible for all d_y cases tested, indicating a minimum effect on the front propeller thrust when placing the rear propeller beneath the front propeller. This is also reflected in the power coefficient (C_p) shown in Figure 7. a b and c, where the difference in C_p between different cases is also negligible.

Recall from Figure 5, that the rear propeller operates at a higher rotational speed than the front propeller for trim. In this case, at the same V_∞ , the resulting J for the rear is lower than for the front propeller. As the changes in thrust or power due to rotational speed or Reynolds number are less than 3%, we can ascribe measured changes in propeller performance to the interference between the two. C_T vs. J and d_y is shown in Figure 6. d e and f. Similar to the front propeller results, C_T is fairly constant until $J \sim 0.1$. However, there is a clear reduction in the rear propeller C_T , for $J > 0.15$; C_T is as much as 30% lower than for the standalone propeller at the same J , particularly in the range of $0.15 < J < 0.3$. Similar to the sudden increase in the rear propeller rotational speed seen in Figure 5, the reduction in C_T occurs at a lower J for a higher propeller vertical spacing. Above $J = 0.3$, C_T increases with J , as for a standalone propeller. At higher J , the C_T recovery depends on the vertical spacing of the propellers, but never exceeds the standalone propeller performance. A larger d_z separation results in a smaller interference at higher forward flight speed. Moreover, the slope of C_T with J (dC_T/dJ) for all cases – front propeller, or rear - is almost the same as that of the standalone propeller at $J \geq 0.4$. This suggests that wake interference at higher advance ratio weakens. Overall, the rear propeller experiences a thrust loss between 12% to 30%. When the propeller's horizontal spacing d_y is reduced, the advance ratio at which C_T abruptly decreases and reaches its minimum value for each configuration around $J \sim 0.2$, with a smaller d_y resulting a lower J where this phenomenon occurs. However, the overall rear propeller seems not to be significantly affected by the streamwise overlap, a trend also seen for the front propeller.

Figure 7. d e and f present the C_p for the rear propeller corresponding to the cases depicted in Figure 6. d e and f. A sudden reduction in C_p is also observed at the same J range where the sudden change in C_T is observed in Figure 7, C_p then recovers at higher J . The maximum C_p difference between the rear propeller and the standalone propeller is $\sim 5\%$, which is much smaller than the loss in thrust generation. In this case, the horizontal and vertical propeller spacing on the C_p is almost negligible. Hence, the analysis of the changes in both C_T and C_p is required for further understanding of the overall tandem propeller system performance, which will be discussed next.

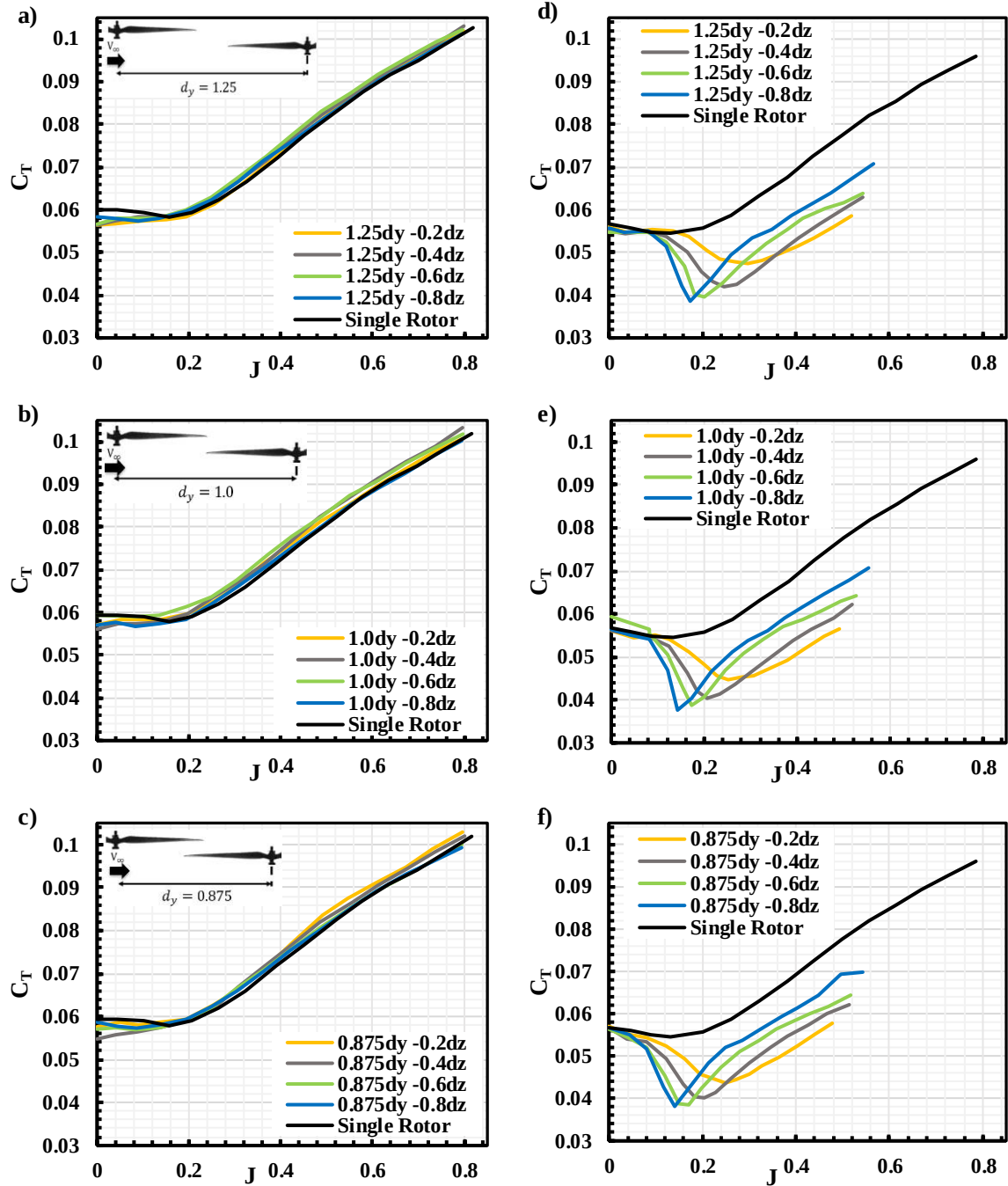


Figure 6. Front Propeller (a-c) and Rear Propeller (d - f) C_T vs. J and d_z , at $\theta = 90^\circ$ and $d_y = 1.25$, $d_y = 1.0$ and $d_y = 0.875$.

A. 2) Tandem Propeller System Performance and Power Consumption at $\theta = 90^\circ$

For tandem propeller systems, the key metric is not the thrust or power of each propeller individually, but rather the efficiency of total thrust production. The optimal arrangement of propeller streamwise and vertical spacing, taking into account packaging constraints, is one that minimizes power consumption while delivering the required thrust. To better represent this, the power required for the propeller pair (P_t) at each (d_y, d_z) and V_∞ is compared with the power required of two standalone single propellers (P_s) generating the same total thrust at the same respective V_∞ . The ratio of P_t and P_s , denoted by $P_{t/s}$, represents the overall system power consumption of the tandem configuration when compared to two single propellers.

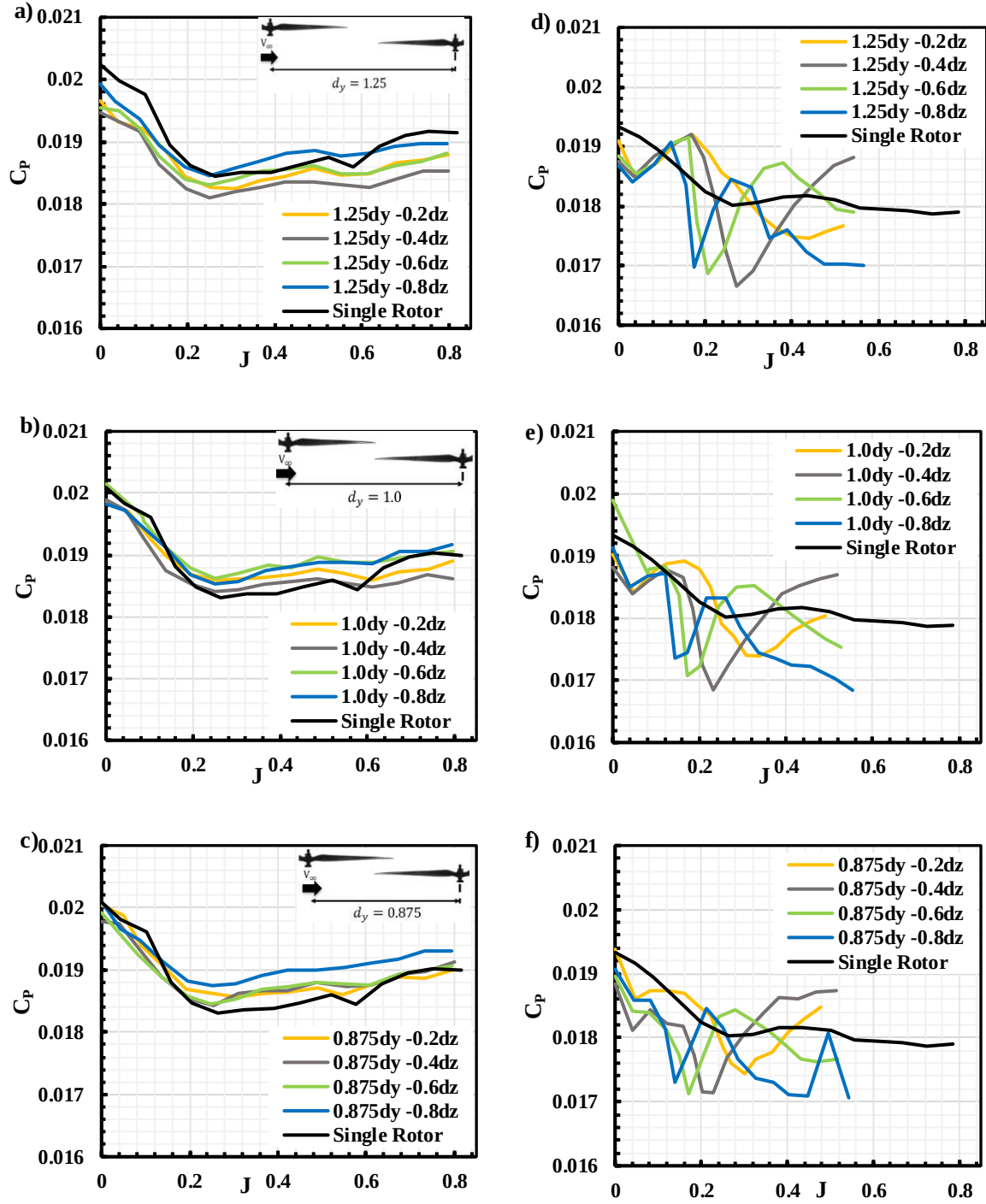


Figure 7. Front Propeller (a-c) and Rear Propeller (d - f) C_p vs. J and d_z at $\theta = 90^\circ$ and $d_y = 1.25$, $d_y = 1.0$ and $d_y = 0.875$.

To calculate P_t/P_s , the sum of the thrust (T_t) and power (P_s) for both tandem propellers is calculated at each V_∞ . Then, using the single propeller data from our previous study [21], between 67 and 117 RPS at the same V_∞ , the rotational speed of the single propeller that generates $T_s = T_t/2$ is found, along with the corresponding power consumption at this rotational speed, forming P_s . $P_t/P_s = 1$ is the ideal or zero interference. Any adverse interference between the propellers means $P_t/P_s > 1$, while $P_t/P_s < 1$, which is not physically likely, would be a favorable interaction that somehow exceeds the efficiency of two propellers in isolation. Thus, the practical goal of a tandem propeller arrangement would be to drive P_t/P_s as close as possible to 1.0, subject to space constraints,

across the range of advance ratios relevant to flight. As the front propeller rotational speed is fixed, while the rear propeller rotational speed varies, the advance ratio is calculated with respect to the front propeller.

Figure 8 shows P_t/P_s vs. J and d_z at $\theta = 90^\circ$. The different colors in Figure 8 correspond to the vertical separation d_z , and the symbols to different streamwise spacings, d_y . In general, the variations in Figure 8 are larger between the various colors than the symbol shapes, or in other words, vertical spacing matters more than streamwise spacing. P_t/P_s remains ~ 1.0 at $J < 0.1$, as expected from Figure 6. Between $0.1 < J < 0.4$, P_t/P_s increases for all cases, as a result of the reduction in C_T for the rear propeller in this advance ratio range. For higher propeller vertical spacing of $d_z < -0.4$, the peak P_t/P_s is reached at this J range due to the higher loss in C_T for the rear propeller seen in Figure 6. At $J > 0.4$, changes in P_t/P_s with J become negligible for $d_z \leq -0.4$. This advance ratio boundary increased to $J = 0.6$ for $d_z = -0.2$, which is very similar to the result seen in our previous study. As the front propeller has a negative effect on the rear propeller at small vertical spacing, P_t/P_s becomes higher for higher d_z at $J > 0.6$. A maximum P_t/P_s of 1.75 is observed at $d_z = -0.2$ at the highest advance ratios. P_t/P_s plateaus to 1.55 for $d_z = -0.4$ and 1.45 for $d_z = -0.6$. For $d_z = -0.8$, the max P_t/P_s is lower than that for $d_z = 0.0$, however, the maximum P_t/P_s for $d_z = -0.8$ occurs at $J \sim 0.25$ where $P_t/P_s = 1.4$, indicating a worse performance at small forward flight speed.

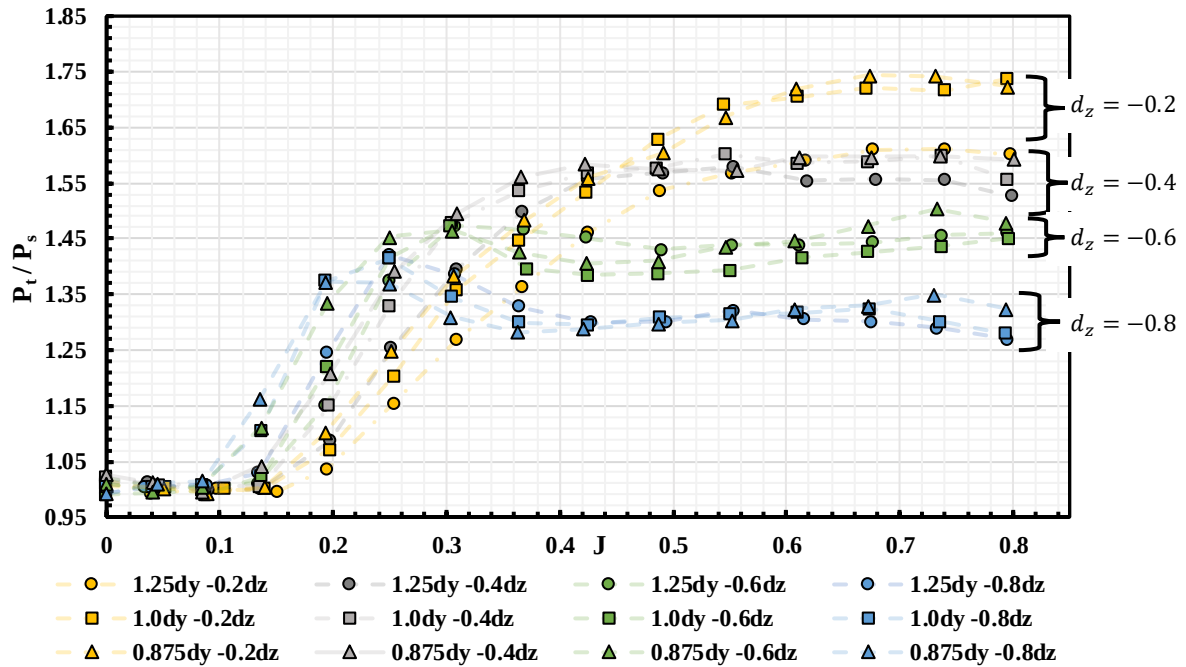


Figure 8. P_t/P_s at $\theta = 90^\circ$ vs. J .

As for the horizontal spacing, the difference is significant at $d_z \geq -0.4$ where a larger d_y results in a lower P_t/P_s , especially for $d_z = -0.2$ where a 9% increment in P_t/P_s is observed when the propeller overlaps. This is to be expected, as the lowest amount of inter-propeller interference should be where the propellers are spaced further apart. Overall, placing the rear propeller beneath the front propeller results in a significant loss in system efficiency when compared to the traditional layout in our previous study [21]. Hence, to improve performance, the rear rotor is required to be placed at least one diameter away from the front rotor vertically, which might increase the overall footprint of the vehicle.

B. Tandem Propeller Configuration at $\theta = 75^\circ$

$\theta = 75^\circ$ is more representative of the multi-copter in cruise, where the vehicle has to pitch nose-down to generate a forward thrust component from the lifting-propellers. Similar to $\theta = 90^\circ$, the rear propeller is operated at a higher rotational speed to achieve pitching moment trim for the tandem propeller system.

B. 1) Effect of Front and Rear Propeller Interaction at $\theta = 75^\circ$

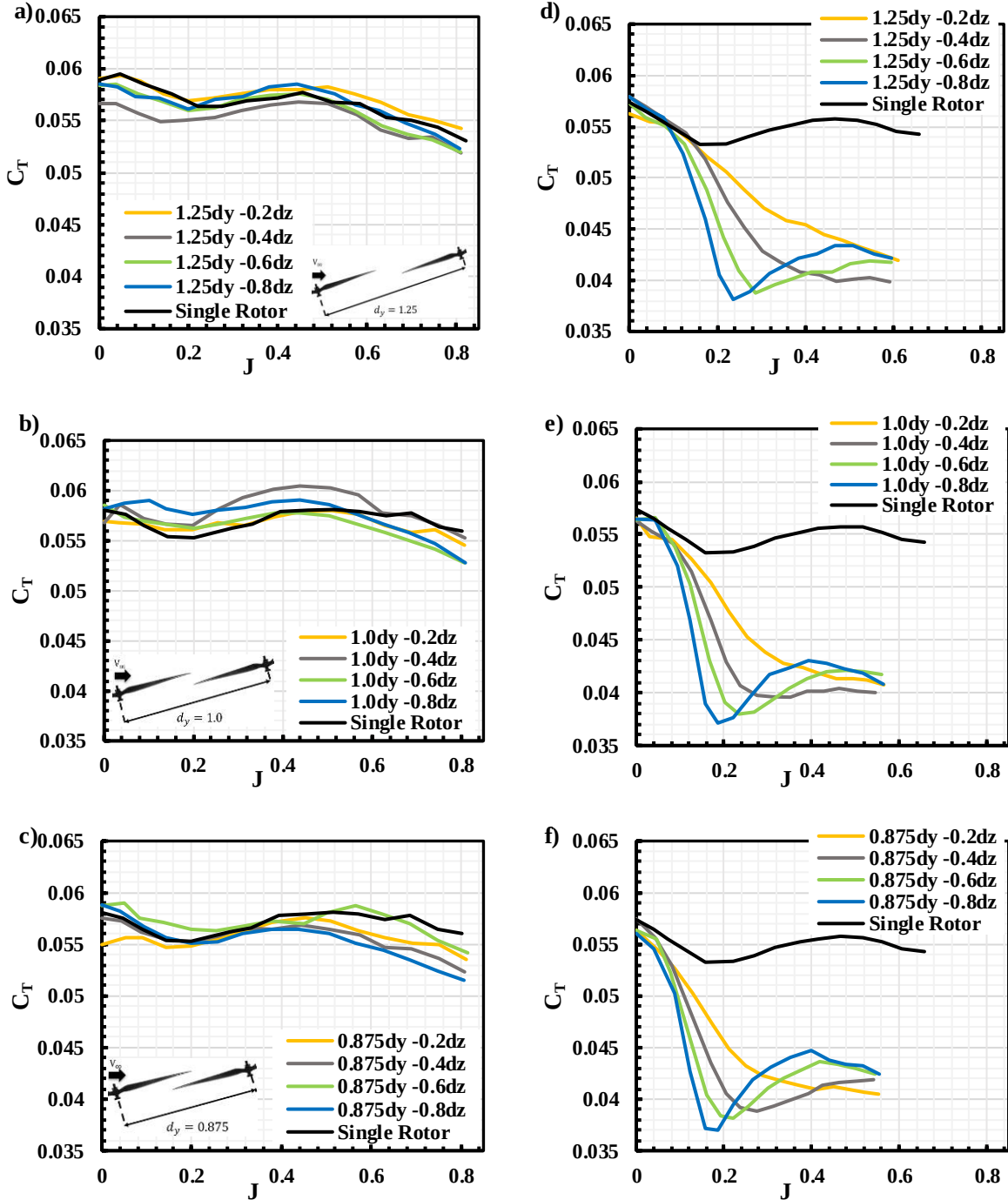


Figure 9. Front Propeller (a-c) and Rear Propeller (d-f) C_T vs. J at $\theta = 75^\circ$ and $d_y = 1.25$, $d_y = 1.0$ and $d_y = 0.875$.

Similar to $\theta = 90^\circ$ in Figure 6 and Figure 7, the effect of the rear propeller on the front propeller is small. C_T and C_P at all cases tested under $\theta = 75^\circ$ are shown in Figure 9 (a-c) and Figure 10 (a-c), respectively. For

all d_z , C_T of the front propeller is almost identical to the standalone propeller, while C_P is slightly higher than that of the standalone case at $J < 0.6$. This agrees with the trend seen in our previous study [21].

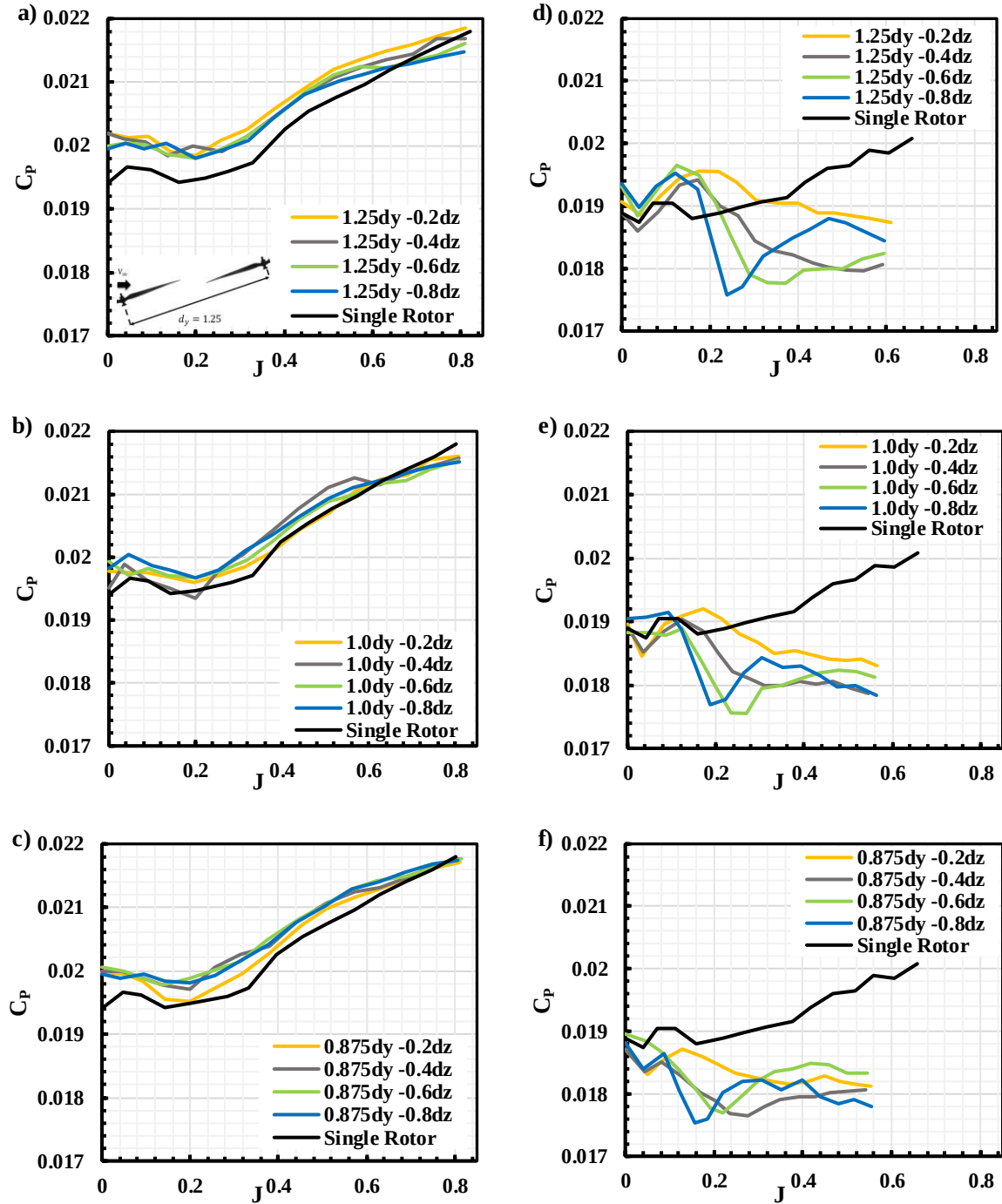


Figure 10. Front Propeller (a-c) and Rear Propeller (d-f) C_P vs. J at $\theta = 75^\circ$ and $d_y = 1.25$, $d_y = 1.0$ and $d_y = 0.875$.

Figure 9 (a-c) shows the rear propeller C_T at different J . Similar to the $\theta = 90^\circ$ cases, the differences in C_T between different vertical and horizontal propeller spacings are negligible at $J \leq 0.1$. At $0.1 < J < 0.4$, variation in C_T across d_z becomes significant. A more negative d_z (higher vertical spacing) results in a higher reduction in C_T , as well as a lower J where this reduction in C_T occurs. Both trends are similar to the results for $\theta = 90^\circ$ in Figure 6. Moreover, when reducing d_y to where the propellers start to overlap, the reduction in C_T occurs at a lower J for the same d_z when compared to the non-overlapping cases, which is also similar to

the $\theta = 90^\circ$ cases. However, at $J > 0.4$, C_T curves merge for all negative d_z cases. C_T for all cases never approaches the standalone propeller, as already observed at $\theta = 90^\circ$. Similar to the results in Figure 6, dC_T/dJ for all rear propeller cases is almost the same as that of the standalone propeller or front propeller at $J \geq 0.4$.

C_P vs. J are shown in Figure 9 (d-f). The overall trend resembles $\theta = 90^\circ$ in Figure 7, where a reduction in C_P is observed with the reduction in C_T at $J < 0.3$ for all $d_z \leq -0.4$ cases. At $J > 0.3$, the deviation between the standalone propeller and all rear propeller cases increases, which is similar to the trend observed in C_T . Similarly, the variation of C_P decreases when overlapping the propellers. The trend in C_P further supports the hypothesis that the front propeller has a constant interference effect on the rear propeller in forward flight.

B. 2) Tandem Propeller System Performance and Power Consumption at $\theta = 75^\circ$

P_t/P_s is calculated for the $\theta = 75^\circ$ cases as in Figure 8, and is shown in Figure 11. Different from the $\theta = 90^\circ$ cases, the variation in P_t/P_s due to vertical spacing (color of symbols in Figure 11) has a similar effect as the horizontal spacing (symbols shape in Figure 11). But the maximal increase in P_t/P_s for $\theta = 75^\circ$ is only about half of that for $\theta = 90^\circ$, indicating that the interference effect of tandem propellers at $\theta = 75^\circ$ is smaller than for $\theta = 90^\circ$. This was also observed in our previous study [21] for negative stagger (rear propeller above the front).

Similar to $\theta = 90^\circ$, a larger vertical spacing result in a steeper increase in P_t/P_s at J just above 0.1. P_t/P_s reaches its maximum between $0.25 < J < 0.5$, which is a common advance ratio range for cruise conditions [13, 15-17]. Further increase in J results in a slight decline in P_t/P_s , with the various cases merging at $J > 0.6$ to $P_t/P_s \sim 1.2$. In summary, when inclining the tandem rotor in forward flight, the rotor vertical spacing becomes less important than in purely edgewise flight ($\theta = 90^\circ$). The interference effects in the inclined configuration attenuate at a high advance ratio. The implication for vehicle design is to avoid inclined flight at intermediate advance ratios.

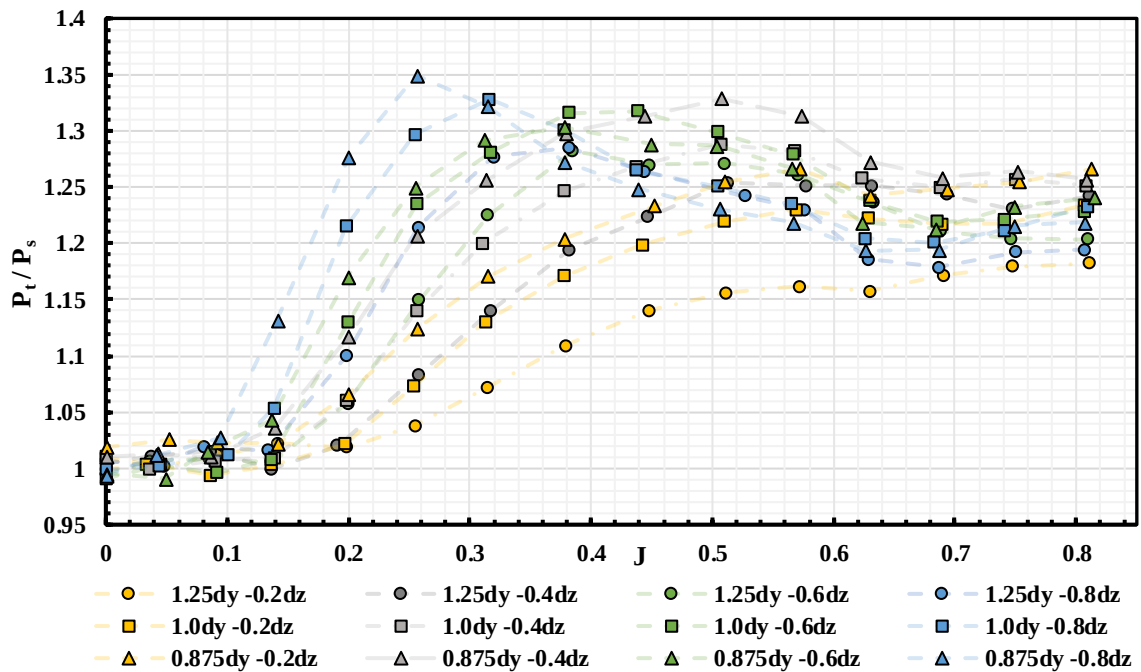


Figure 11. P_t/P_s at $\theta = 75^\circ$ vs. J .

C. Smoke Flow Visualization

Flow visualization is conducted for $d_y = 1.25$, $d_z < 0$ and $\theta = 90^\circ$. From the rear propeller C_T (Figure 6) and the tandem system P_t/P_s (Figure 8), the advance ratio where the rear propeller reaches its minimum C_T and the tandem system reaches its maximum P_t/P_s are slightly different at different d_z . Increasing the vertical displacement results in both extremes occur at a lower advance ratio than for smaller vertical displacement.

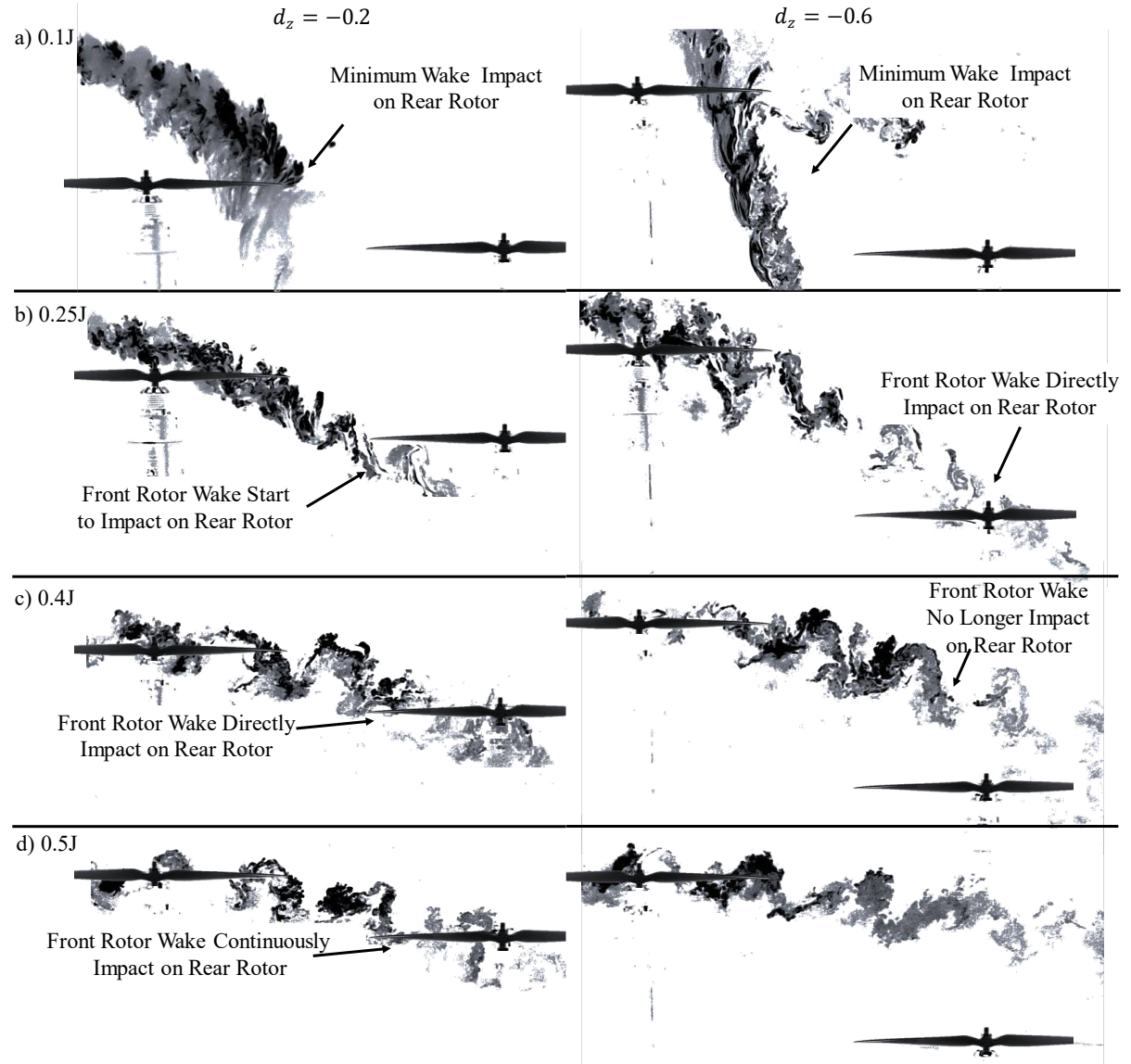


Figure 12. Flow visualization for $d_y = 1.25$ and $d_z = -0.2$ and -0.6 at (a) $J = 0.1$, (b) $J = 0.25$, (c) $J = 0.4$ and (d) $J = 0.5$.

Figure 12 shows that at $J = 0.1$, the wake of the front propeller did not impact on rear propeller either for a small vertical spacing ($d_z = -0.2$) or a large one ($d_z = -0.6$). It is not surprising that thrust and power for the rear propeller, the front, or the tandem pair, are indistinguishable from that of a single propeller in isolation. At $J = 0.25$ (rear rotor equivalent J of ~ 0.19), the wake of the front propeller directly impacted the rear propeller at $d_z = -0.6$, while at $d_z = -0.2$ it just started to impact the rear propeller. This agrees with the C_T results where C_T of the rear propeller at $d_z = -0.6$ reaches its minimum at $J = 0.25$, while for $d_z = -0.2$ at $J = 0.25$, C_T just started to deviate from the single propeller's curve. On the other hand, P_t/P_s for $d_z = -0.6$

almost reaches its maximum, while for $d_z = -0.2$ it just starts to rise from 1.0. At $J = 0.4$ (rear rotor equivalent J of ~ 0.3), now with higher freestream momentum, the wake of the front propeller at $d_z = -0.6$ no longer directly impacts the rear propeller, while for $d_z = -0.2$ it directly impacts the rear propeller. Similarly, the rear propeller C_T reaches its minimum, and the tandem propeller P_t/P_s reaches its maximum. A further increase in J has a relatively small effect on the front propeller wake angle, as the freestream momentum already dominates the propeller wake momentum. This correlates with constant dC_T/dJ (Figure 6), matching that of the single propeller. The tandem system P_t/P_s also no longer varies with advance ratio (Figure 8).

V. Conclusions

Forces and moments were measured for a tandem pair of propellers across a range of advance ratios in an open-jet wind tunnel. The rear propeller was placed beneath the front propeller, with the pair at 90-degree incidence with respect to the free stream, or purely edgewise flight; and at 75-degree incidence, to represent the cruise attitude of a conventional multi-copter. The efficiency of the tandem pair was referenced to that of a propeller in isolation.

Placing the rear propeller beneath the front propeller results in a significant decrease in the efficiency of the rear propeller, due to wake interaction. This loss occurs at a lower advance ratio when increasing the propeller vertical spacing. For a higher advance ratio, a large propeller vertical spacing might not lead to a higher loss in the overall system performance as the wake of the front propeller passes above the rear propeller. Reducing the propeller vertical spacing at a higher advance ratio results in a reduction in the system performance in pure edgewise flight, especially for $d_z = -0.2$, where the wake of the front propeller continuously impacts the rear propeller. However, when tilting the tandem system forward by fifteen degrees, the trend is not preserved. A higher vertical spacing does not bring a higher performance benefit, as a result of the propeller wake interaction. Increasing the horizontal propeller spacing shows a consistent small benefit for both incidence angles.

Flow visualization confirms that when the front propeller wakes directly impacts the rear propeller, the efficiency of the rear propeller reduces. This occurs at $0.1 < J < 0.4$. For $d_z < 0.4$. Further increase in J will result in a smaller interference, as the wake from the front propeller no longer directly impacts the rear propeller. However, the front propeller would continuously affect the inflow of the rear propeller as both efficiencies never exceed the standalone propeller performance.

In all cases of purely edgewise flight, after passing an advance ratio threshold where the thrust coefficient is a minimum, thrust increases linearly with advance ratio, at a universal value of slope – the same for a propeller in isolation, the front propeller in a tandem pair, or the rear.

VI. References

- [1] Leishman, J. G., “Principles of Helicopter Aerodynamics”, Cambridge University Press, 2006.
- [2] Dingeldein, R. C., “Wind Tunnel Studies of the Performance of Multirotor Configurations,” NACA TN 3236, August 1954.
- [3] Nagashima, T., O. H., and Sasaki, F., “Optimum Performance and Load Sharing of Coaxial Rotor in Hover,” Journal of Japan Society for Aeronautics & Space Sciences, Vol. 26, No. 293, June 1978, pp. 325–333.
- [4] Lee, T. E., and Leishman, J. G., “Design and Performance of a Ducted Coaxial Rotor in Hover and Forward Flight,” M. S. Thesis, University of Maryland, MD, October 2010.

- [5] Ramasamy, M., “Measurements Comparing Hover Performance of Single, Coaxial, Tandem, and Tilt-Rotor Configurations” AHS 69th Annual Forum, Phoenix, AZ, May 2013
- [6] Otsuka, H., Sasaki, D., and K. Nagatani, “Reduction of the head-up pitching moment of small quad-rotor unmanned aerial vehicles in uniform flow,” *Int. J. Micro Air Veh.*, pp. 1–21, 2017.
- [7] Otsuka, H., and Nagatani, K., “Pitching moment caused by rotor flow interference on a small quad-rotor unmanned aerial vehicle in a uniform flow,” in *Transactions of the Japan Society for Aeronautical and Space Sciences*, 2018, vol. 61, no. 2, pp. 87–96.
- [8] Volocopter, VoloCity air taxi, Retrieved from: volocopter.com.
- [9] Fukushima, Y., and Oyama, A., “Aerodynamic interaction of overlapping rotors for Mars drone” AIAA SciTech 2022, January 2022, AIAA 2022-0731.
- [10] Felker, F. F., and J. S. Light, “Rotor / wing Aerodynamic Interactions in Hover,” 1986.
- [11] Sweet, G. E., “Hovering measurements for twin-rotor configurations with and without overlap,” *Natl. Aeronaut. Sp. Adm.*, 1960.
- [12] Huston, J., “Wind-Tunnel Measurements of Performance, Blade Motions, and Blade Air Loads for Tandem-Rotor Configurations with and without Overlap,” Hampton, Va, 1963.
- [13] Prudden, S. L., “Rotor Aerodynamic Interaction effect for multirotor unmanned aircraft system in forward flight” Ph.D. Thesis, RMIT University, Australia, September 2020
- [14] Tsaltas, J.D., “Trim Solutions of Multirotor Vehicles using a Fast Performance Prediction Method,” Master’s Thesis, Ryerson University, December 2018.
- [15] Atte, A., Wylie, D., and Rauleder, J., “Experimental Evaluation of Multi-Rotor Aerodynamic Interactions”, VFS 78th Annual Forum, May 2022.
- [16] McCrink, M., Seth, D., and Herz, S., “Flight Test Measurement of Quadrotor Performance at Varying Sideslip Angles” VFS 78th Annual Forum, May 2022.
- [17] Herz, S, Atte, A, Seth, D, Rauleder, J, and McCrink, M., “Impact of Rotor–Rotor and Rotor–Body Aerodynamic Interactions on Quadrotor Vehicle Performance”, VFS 77th Annual Forum, May 2023.
- [18] Kostek, A. A., Braukmann, J. N., Loßle, F., Miesner, S., Visingardi, A., Boisard, R., Riziotis, V., Keßler, M., and Gardner, A. D., “Experimental Investigation of Quadrotor Aerodynamics with Computational Cross-Validation” VFS 77th Annual Forum, May 2023.
- [19] Otsuka, H., and Nagatani, K., “Thrust Loss Saving Design of Overlapping Rotor Arrangement on Small Multirotor Unmanned Aerial Vehicles” 2016 IEEE ICRA, May 2016, IEEE-3242
- [20] Weishäupl, A., and Prior, S., “Influence of Propeller Overlap on Large-Scale Tandem UAV Performance”, *Unmanned Systems*, Vol. 07, No. 04, pp. 245-260, May 2019, doi: 10.1142/S2301385019500092
- [21] Cai, J., Gunasekaran, S., and OL, M. V., “Vertically Offset Overlapping Propellers in Tandem Configuration” AIAA SciTech 2023, National Harbor, MD, January 2023, AIAA 2023-2464.
- [22] Cai, J., and Gunasekaran, S., “Sinusoidal Gust Response of RC Propellers Tandem Configuration” VFS Forum 79, West Palm Beach, FL, May 2023, F-0079-2023-11540.
- [23] KDE Direct, KDE CF-125 Propeller, Retrieved from: kdedirect.com

[24] ATI, F/T Mini-40 Sensor, Retrieved from: www.ati-ia.com

[25] MathWorks, Designfilt, FIR band stop filter: <https://www.mathworks.com/>. [retrieved Feb. 2019]